

Student Name:
Student Number:
Period:

Calculus I Exam

Limits, Derivatives, and Integrals
Mr. Angerhofer
Date:

1. Answer the questions in the spaces provided. If you run out of room, ask for scratch paper.
2. Work must be shown for partial credit and is required when a question asks for an explanation.
3. The exam has a no-calculator portion followed by a calculator portion.
4. If you have a question, raise your hand and wait for the teacher.

Total: 62 points

Time: 60 minutes

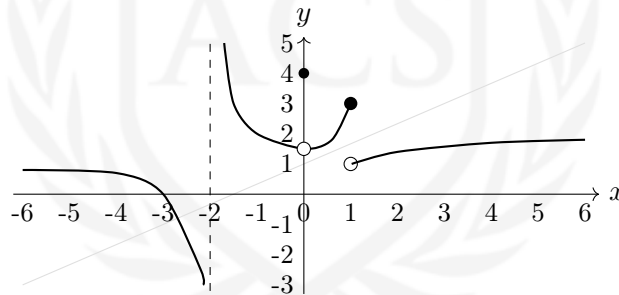
Part I: No Calculator

1. (3 points) Use the table to evaluate the limit. Explain briefly how the table supports your answer.

x	1.9	1.99	2	2.01	2.1
$f(x)$	3.91	3.991	undefined	4.009	4.09

 $\lim_{x \rightarrow 2} f(x) = \underline{\hspace{2cm}}$

2. Use the graph of $f(x)$ below to answer the following questions.



- (a) (1 point) $\lim_{x \rightarrow \infty} f(x) =$
- (b) (1 point) $\lim_{x \rightarrow -2^+} f(x) =$
- (c) (1 point) $\lim_{x \rightarrow -2^-} f(x) =$
- (d) (1 point) $\lim_{x \rightarrow -2} f(x) =$
- (e) (1 point) $\lim_{x \rightarrow 0} f(x) =$
- (f) (1 point) $\lim_{x \rightarrow 1^-} f(x) =$
- (g) (2 points) Is $f(x)$ continuous at $x = 0$? Use the definition of continuity to explain why or why not.

3. Let

$$\begin{cases} f(x) = x^2 - 4, & x \leq 2, \\ 3x - 6, & x > 2. \end{cases}$$

(a) (3 points) Find $\lim_{x \rightarrow 2^-} f(x)$ and $\lim_{x \rightarrow 2^+} f(x)$.(b) (2 points) Is f continuous at $x = 2$? Use the definition of continuity to justify your answer.4. Suppose $\lim_{x \rightarrow c} f(x) = 5$ and $\lim_{x \rightarrow c} g(x) = -2$. Find each limit.(a) (1 point) $\lim_{x \rightarrow c} [2f(x) - 3g(x)]$ (b) (1 point) $\lim_{x \rightarrow c} \frac{g(x)}{f(x)}$ (c) (1 point) $\lim_{x \rightarrow c} [f(x)g(x)]$

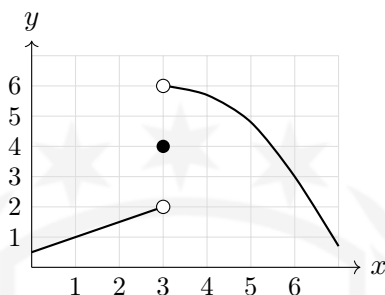
5. Find each limit algebraically, if it exists.

(a) (2 points) $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$ (b) (2 points) $\lim_{x \rightarrow \infty} \frac{6x^2 - x + 1}{2x^2 + 5}$

6. (3 points) Find the x -values, if any, at which $f(x)$ is not continuous.

$$f(x) = \begin{cases} x^2 - 2, & x \leq 1, \\ \frac{x^2 - 4}{x - 2}, & x > 1. \end{cases}$$

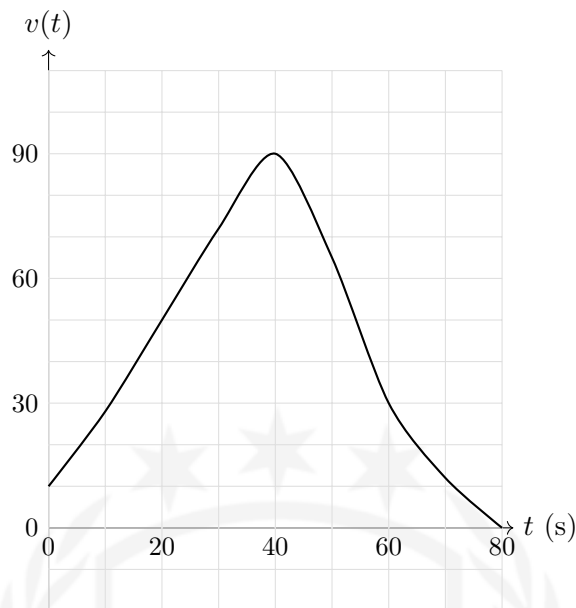
7. (2 points) Which of the following expressions equals 2 for the graph shown below?



- (A) $f(3)$ (C) $\lim_{x \rightarrow 3^+} f(x)$
 (B) $\lim_{x \rightarrow 3^-} f(x)$ (D) $\lim_{x \rightarrow 3} f(x)$
8. (2 points) For $x < 0$, the line $y = 3$ is an asymptote of the graph of a function f . Which statement must be true?
- (A) $f(x) \neq 3$ for $x < 0$ (C) $\lim_{x \rightarrow \infty} f(x) = 3$
 (B) $\lim_{x \rightarrow -\infty} f(x) = 3$ (D) $\lim_{x \rightarrow 3} f(x) = \infty$
9. (2 points) If f is continuous on the closed interval $[2, 5]$, $f(2) = -3$, and $f(5) = 4$, which statement must be true?
- (A) $f(c) = 2$ for some $c \in (-3, 4)$ (C) $f(c) = 0$ for some $c \in (-3, 4)$
 (B) $f(c) \neq 1$ for every $c \in (2, 5)$ (D) $f(c) = 2$ for some $c \in (2, 5)$

Part II: Calculator Allowed

10. The velocity $v(t)$, in feet per second, of a car is represented by the graph below.



- (a) (2 points) Estimate the instantaneous rate of change of v at $t = 60$ by drawing a tangent line and using a difference quotient.
- (b) (1 point) What does the rate of change in part (a) represent in this context?
- (c) (2 points) Estimate the distance traveled from $t = 20$ to $t = 60$ by counting geometric areas.
- (d) (3 points) Estimate the total distance traveled from $t = 0$ to $t = 80$ by using the Trapezoid Rule with four subintervals.

11. (5 points) For $s(t) = -t^2 + 12t + 5$, find the average velocity on the interval $[2, 2.1]$. Then state the instantaneous velocity at $t = 2$ by taking the appropriate limit.

12. A particle moves along a line with position

$$s(t) = 0.4t^3 - 3t^2 + 8t + 2$$

for $0 \leq t \leq 6$, where s is measured in meters and t in seconds.

- (a) (2 points) Find the particle's average velocity on $2 \leq t \leq 5$.
- (b) (2 points) Use the calculator to find the instantaneous velocity at $t = 3$, and interpret its sign.

13. The graph of f is modeled by

$$f(x) = \frac{x^2 - 5x + 6}{x - 2}.$$

- (a) (2 points) Use algebra to identify any discontinuity(s).
- (b) (2 points) Use a graphing calculator to display the graph near the discontinuity and state the value of $\lim_{x \rightarrow 2} f(x)$.
- (c) (1 point) Find the horizontal asymptote, if one exists.
14. (4 points) Find a and b so that the function is continuous for all real x :

$$f(x) = \begin{cases} 2x + a, & x < -1, \\ ax + b, & -1 \leq x \leq 3, \\ 4x - b, & x > 3. \end{cases}$$

Show the two equations you use and solve the resulting system.

15. (4 points) Verify that the Intermediate Value Theorem applies to $p(x) = x^3 - 4x + 1$ on $[0, 2]$. Use a calculator to locate one value of c in $(0, 2)$ for which $p(c) = 0$, rounded to three decimal places.